WORK-STUDY FROM SEPTEMBER 2026 · POWER ELECTRONICS & ELECTRICAL ENERGY SYSTEMS

Khalil LAGHA

M2 student - Electrical Energy Systems Design (CSEE) · Université Grenoble Alpes

Ranked 2nd of the M1 EEA cohort (15.43/20). DC-DC (buck, boost) and AC-DC converters, regulation-loop tuning (PI/PID), efficiency optimization, EMI compliance: validated in simulation then on real hardware, across five internships from the lab (GIPSA-lab) to industrial sites. Rotation: 2–3 weeks company / university.

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- Authorized to work in France

01 Experience

Research Intern · GIPSA-lab, Grenoble — 04/2026 - 07/2026 · 3.5 months

- Modeled a **10-cell PEMFC fuel-cell stack** in Matlab/Simulink: **6%** dynamic and **1%** static error against real measurements.
- Tuned a **DC-DC boost converter** PI **regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench.

Intern · SLB (Schlumberger) — 07/2025 · 1 month

- Authored technical syntheses **in English** (directional drilling) in an international industrial environment.

Power Electronics & Automation Intern · EURL Lagha — 01/2024 · 15 d + 09/2024 · 1 month

- Contributed to an **industrial DC-DC buck converter**: 50 V input, adjustable 0–48 V output, **25 A**.
- Contributed to **4 transformer-rectifiers** feeding pipelines (electrolysis): three-phase 400 V → 100 V transformer, rectification + capacitive filtering, **100 V DC / 80 A (≈ 8 kW)** output, ripple < 5%, EMI compliance.
- Co-designed the supervision HMI (Siemens TIA Portal, Ladder PLC); analyzed a customer specification, sized the corresponding transformer and produced the electrical schematics in **EPLAN**.

Power Grid Intern · Sonelgaz — 03/2024 · 15 d · observation

- Analyzed the full generation → distribution chain (**EHV 400/220 kV → HV 60 kV → MV 30/10 kV → LV 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit.

02 Projects

Power grid analysis · Matlab · solo
Load flows from **3 to 118 buses** (PQ/PV/slack; IEEE 14/30/57/118 networks); benchmarked **3 numerical methods**: Gauss-Seidel, Newton-Raphson, fast-decoupled (accuracy vs computation time); studied **transient stability.**

Field-oriented control (FOC) of an induction motor · Matlab/Simulink
Implemented Park transforms, PID controllers and observers; tuned the **cascaded current- and speed-regulation loops.**

Electrical machine sizing · Python (OOP, PyQt) · team
Stator winding design, flux density via reluctance networks, finite-element validation (**FEMM**), automatic parameter generation.

Neural network from scratch (MLP) · Matlab · open source (MIT)
Multilayer perceptron and backpropagation hand-coded, no toolbox; debugged, tested and published: github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Also: cascade control of a DC motor (three-phase thyristor bridge, Simulink) · Siemens Step7 automation (HMI + Ladder) · Arduino line-follower robot (Poly Maze competition).

2nd	15.43	C2
of the M1 EEA cohort (UGA)	M1 annual average /20	English

03 Education

M2 CSEE - Electrical Energy Systems Design — from 09/2026
Université Grenoble Alpes · work-study

M1 EEA (EECS) — 2025 – 2026
Université Grenoble Alpes · 2nd of the cohort, annual average 15.43/20

Engineering cycle, Electrical Engineering — 2021 – 2025
École Nationale Polytechnique, Algiers · 4 of 5 years completed; prep: 16th/358 then 49th/264

Baccalauréat in Mathematics — 2021
Highest honours (16.93/20)

04 Skills

Power electronics · DC-DC / AC-DC conversion · Machines & transformers · Grids: load flow, stability · PID/LQR loops · Kalman · FOC · PLC / HMI / SCADA · Electrical CAD: EPLAN, AutoCAD · AI: NN, GA, fuzzy logic

05 Software

Matlab/Simulink (Simscape Power Systems) · Python (OOP, PyQt) · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 (Ladder/SCL) · FEMM · SolidWorks · Arduino · Git/GitHub

06 Languages & certification

French	C1
English	C2

SLB **NEST & SIPP Level 2** · 07/2025 (HSE)

07 Soft skills

Fast learner · Rigorous · Organized · Problem solving · Comfortable with tools

Portfolio: **khalilellahlagha.github.io** · IEEE Student Branch ENP · VIC